

SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No: 7

Aim: Connect to different data sources (Excel, CSV, Database)

- a) Create and use data extracts vs live connections
- b) Manage metadata (rename, hide, change data types)
- c) Perform joins (inner, left, right) on multiple tables
- d) Apply data blending between two sources
- e) Implement different types of filters (dimension, measure, date)
- f) Optimize performance using extracts and filtering

a) Create and use data extracts vs live connections

- In the top-right corner of the Data Source tab, locate the connection mode toggle.
- Select Extract to save a local copy for performance, or keep it on Live for real-time updates.

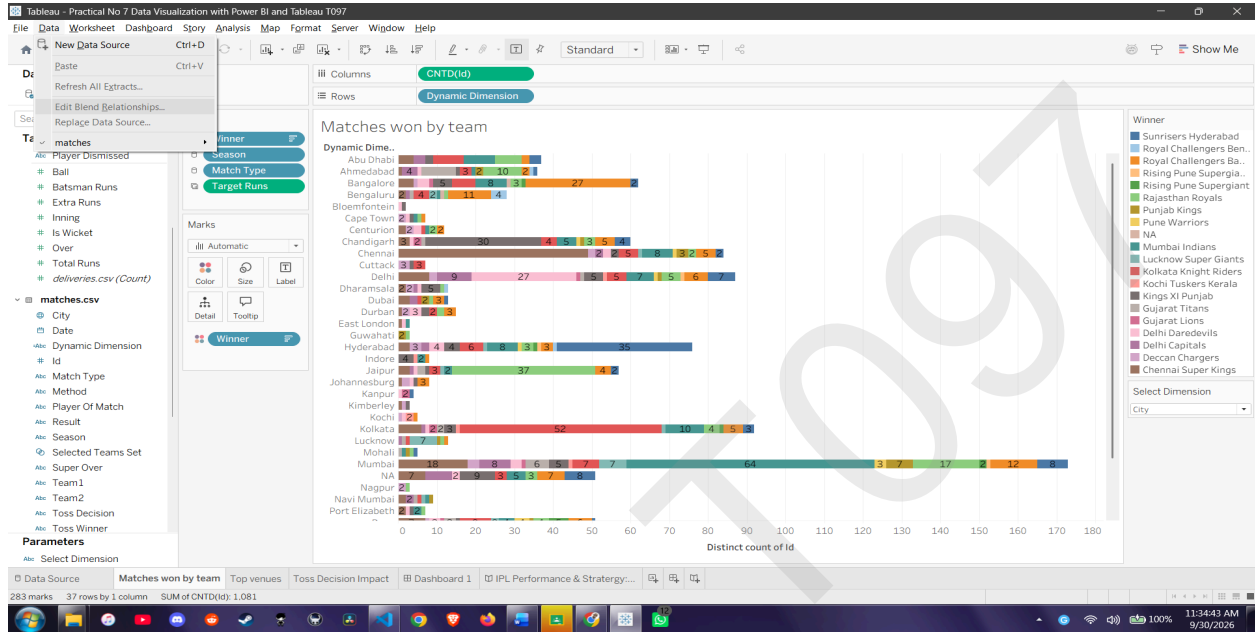
The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane shows 'matches' as the selected data source. Below it, the 'Files' pane lists 'deliveries.csv' and 'matches.csv'. The main workspace displays a relationship diagram with 'matches.csv' and 'deliveries.csv' connected by a line. The bottom pane shows a data table with the following columns: Match Id, Inning, Batting Team, Bowling Team, Over, Ball, Batter, and Bowler. The table contains 6 rows of data, all with Match Id 548335. The 'Inning' column has values 1, 1, 1, 1, 1, 1. The 'Batting Team' column has values Deccan Chargers, Deccan Chargers, Deccan Chargers, Deccan Chargers, Deccan Chargers, Deccan Chargers. The 'Bowling Team' column has values Kolkata Knight Riders, Kolkata Knight Riders, Kolkata Knight Riders, Kolkata Knight Riders, Kolkata Knight Riders, Kolkata Knight Riders. The 'Over' column has values 10, 10, 10, 10, 11, 11. The 'Ball' column has values 3, 4, 5, 6, 1, 2. The 'Batter' column has values PA Patel, S Dhawan, PA Patel, PA Patel, S Dhawan, PA Patel. The 'Bowler' column has values L Balaji, L Balaji, L Balaji, L Balaji, R Bhatia, R Bhatia.

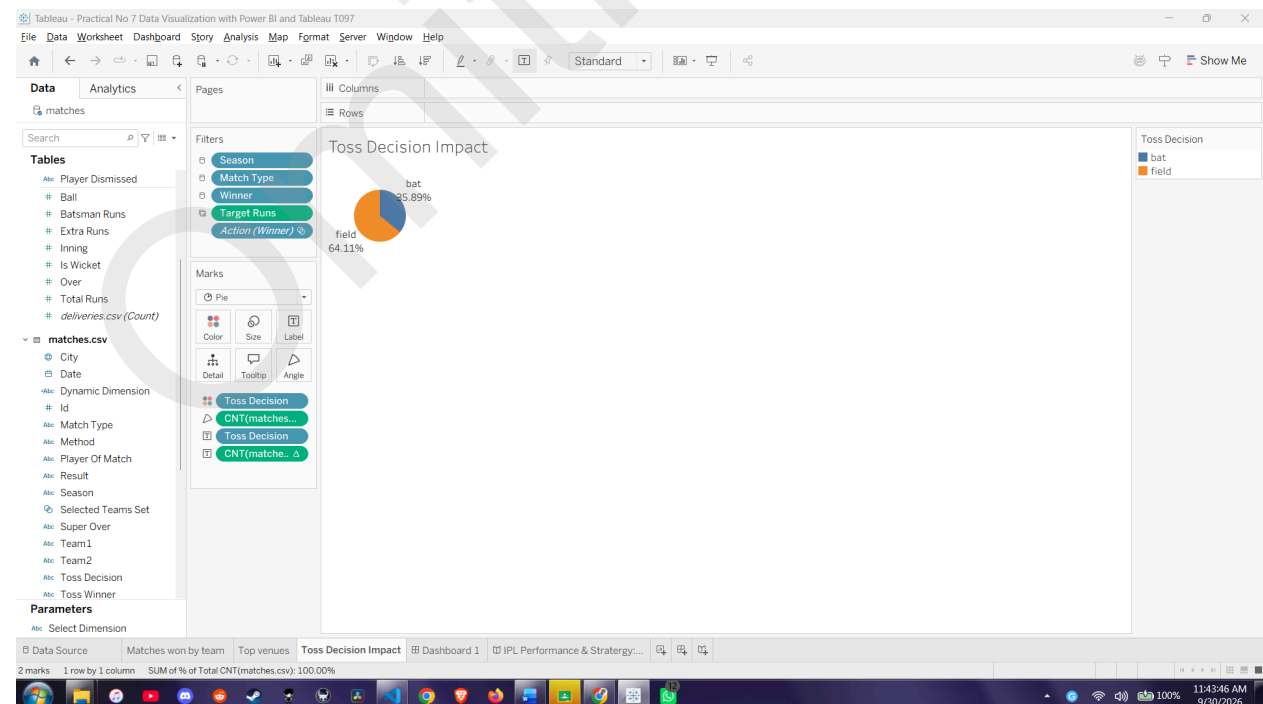
Match Id	Inning	Batting Team	Bowling Team	Over	Ball	Batter	Bowler
548335	1	Deccan Chargers	Kolkata Knight Riders	10	3	PA Patel	L Balaji
548335	1	Deccan Chargers	Kolkata Knight Riders	10	4	S Dhawan	L Balaji
548335	1	Deccan Chargers	Kolkata Knight Riders	10	5	PA Patel	L Balaji
548335	1	Deccan Chargers	Kolkata Knight Riders	10	6	PA Patel	L Balaji
548335	1	Deccan Chargers	Kolkata Knight Riders	11	1	S Dhawan	R Bhatia
548335	1	Deccan Chargers	Kolkata Knight Riders	11	2	PA Patel	R Bhatia

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d) Apply data blending between two sources

- Go to Data → New Data Source to load a completely independent dataset.
- Go to Data → Edit Blend Relationships and map the common linking fields between your Primary and Secondary data sources.



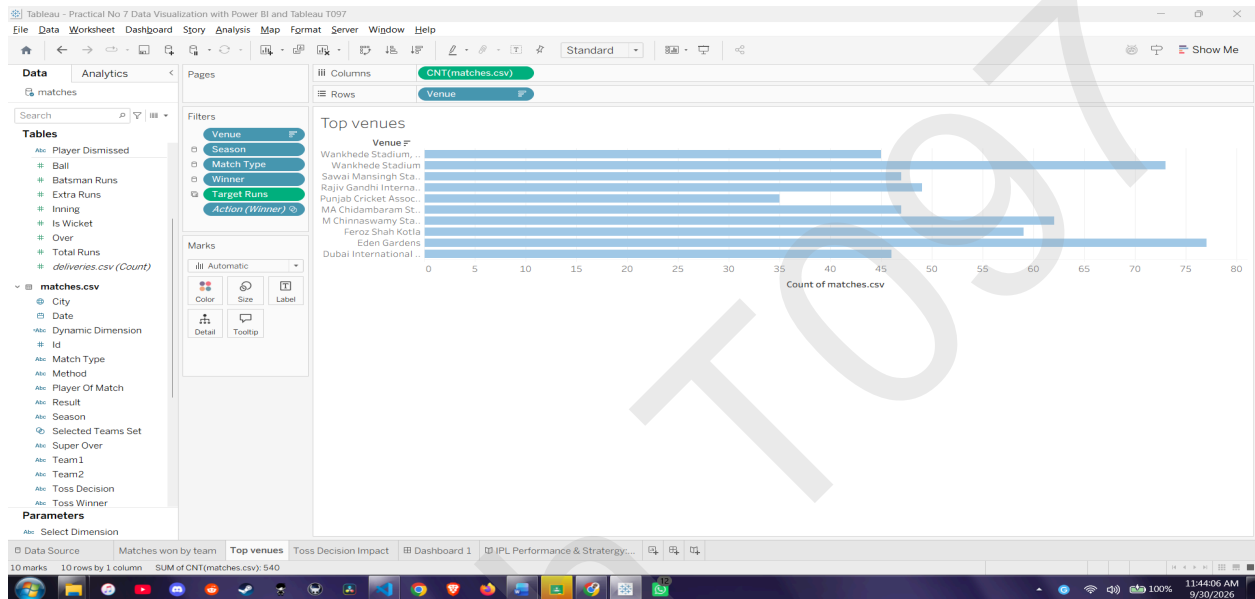


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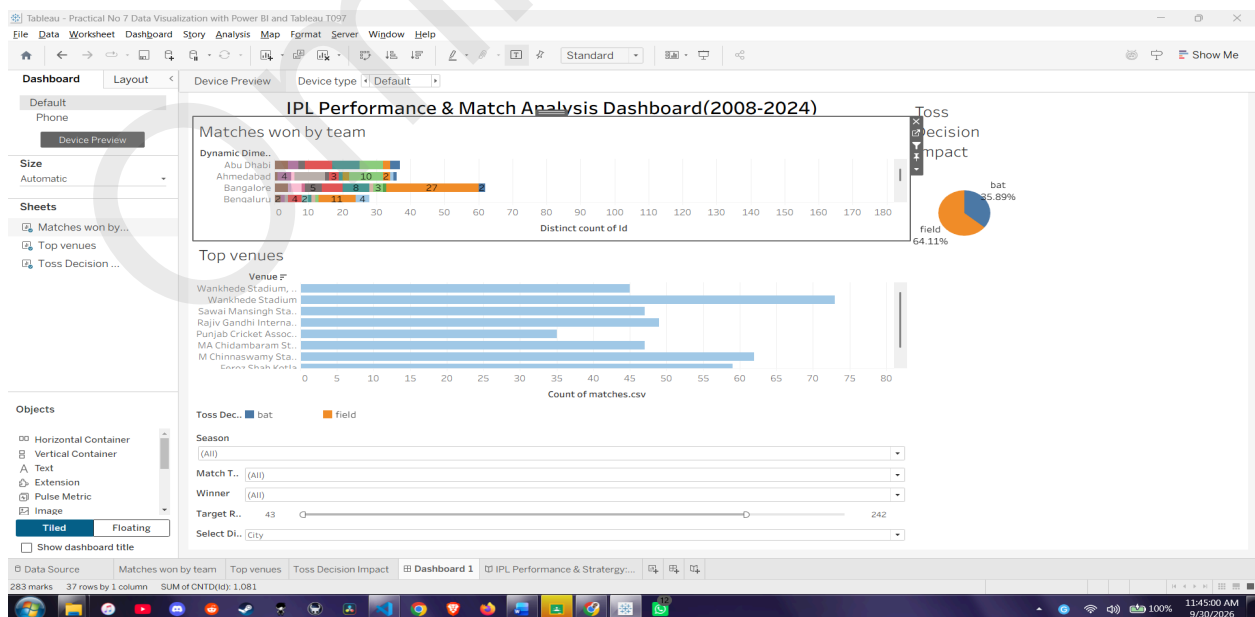
f) Optimize performance using extracts and filtering

- Open the Top venues sheet.
- Right-click the **Venue** dimension pill on your Filters shelf and select **Add to Context**.
- The pill will turn dark gray, forcing Tableau to process this filter first to optimize performance.



Consolidating the Analysis (Dashboards & Stories)

- Combine your worksheets into **Dashboard 1** to create an interactive overview.



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- Create a Story named **IPL Performance & Strategy...** and build sequential story points (e.g., "League Performance Overview", "Venue Dynamics", "Toss Decisions") to present your insights.

